

```

fit_data_path = 'e:\SHARE\FE\Data\analysis\06_fit_with_J0\Sym4D_cutsAndFits';
if ~exist('fit_resEi400','var')
    %fit_data_file = 'FitEn_cut200_1Dsbg_dirGH_dE20_fix_bg_slope.mat'; % fit 1D
background with no Q-slope
    %fit_data_file =
'FitEn_cut200_1D_fit_bg_dirGH_dE20_fix_bg_slope.mat';%Externaly 2-exponent fit
fixed bg at high momentum.
    fit_data_file =
'FitEn_cut400ref110_2D_dirGH_dE20_fix_bg_slope.mat';%Externaly 2-exponent fit
fixed bg at high momentum.
    ld = load(fullfile(fit_data_path,fit_data_file));
    if isfield(ld,'fit_resEi400')
        fit_resEi400 = ld.fit_resEi400;
    else
        fit_resEi400 = ld.fit_res;
    end
end
data_path = 'e:\SHARE\FE\Data\analysis\06_fit_with_J0\sym4D_cutsAndFits';
fit_file = 'multicuts_fit_dataGH_ei400meV.mat';
if ~exist('cuts2fit400','var')
    ld = load(fullfile(data_path,fit_file));
    cuts2fit400 = ld.cuts2fit400;
end

mi = maps_instrument(400,600,'S');
sample=IX_sample(true,[1,0,0],[0,1,0],'cuboid',[0.04,0.03,0.02]);
cl = cuts2fit400.cutsF_list;
for i=1:numel(cl)
    cl{i} = cl{i}.set_instrument(mi);
    cl{i} = cl{i}.set_sample(sample);

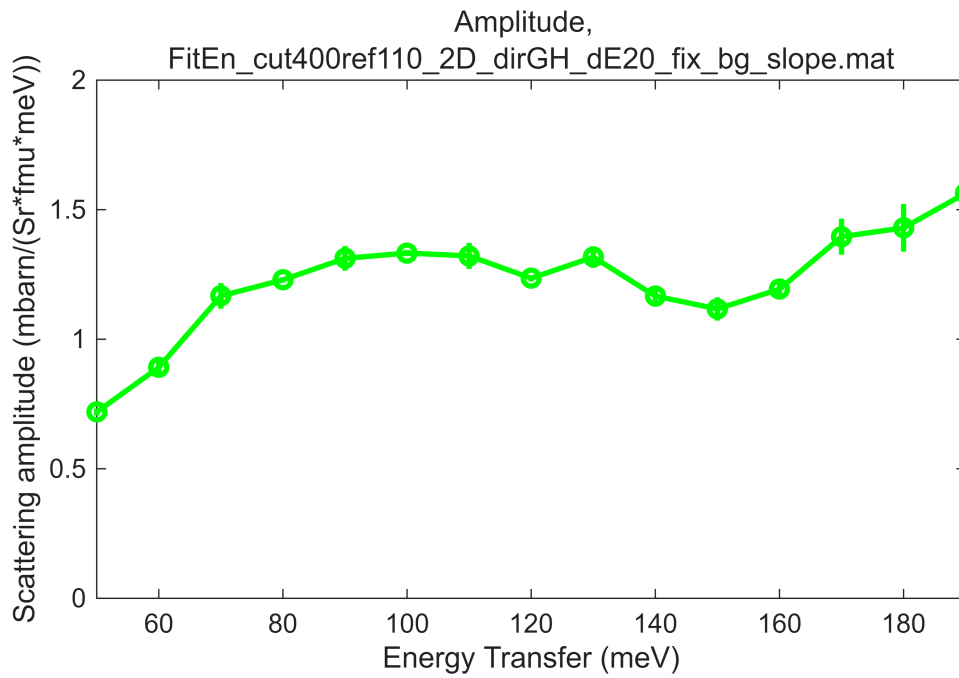
    %cuts2fit.cutsS_list{i} = cuts2fit.cutsS_list{i}.set_instrument(mi);
    %cuts2fit.cutsS_list{i} = cuts2fit.cutsS_list{i}.set_sample(sample);
end
cuts2fit400.cut110_sel = cl{1};
cuts2fit = cuts2fit400;
fit_res = fit_resEi400;

```

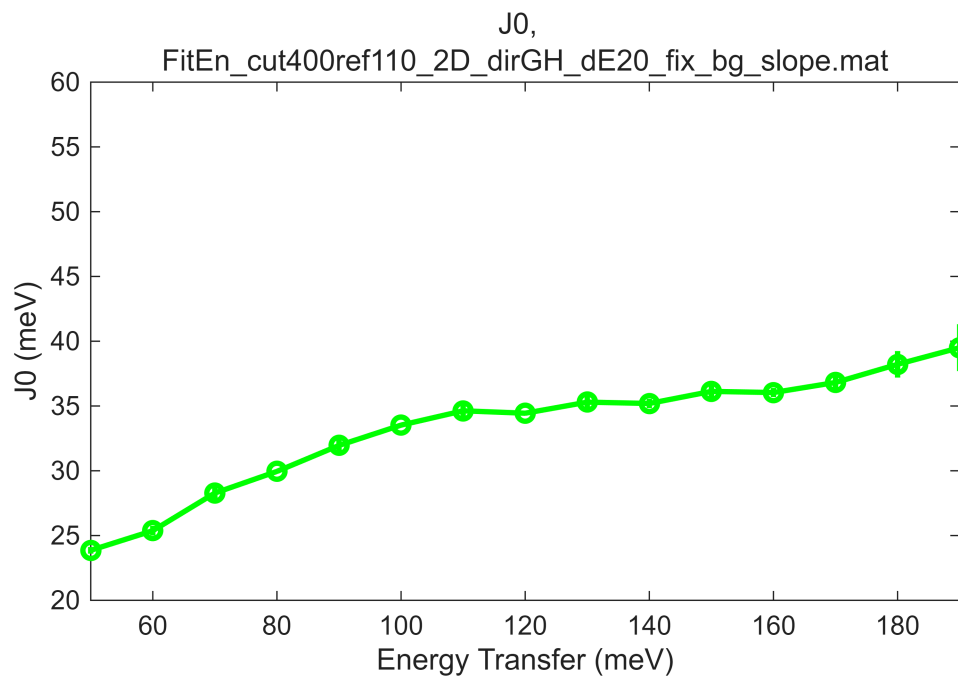
```

acolor g
Sgn1 = fit_res.S;
pd(Sgn1);ly 0 2;liny;keep_figure

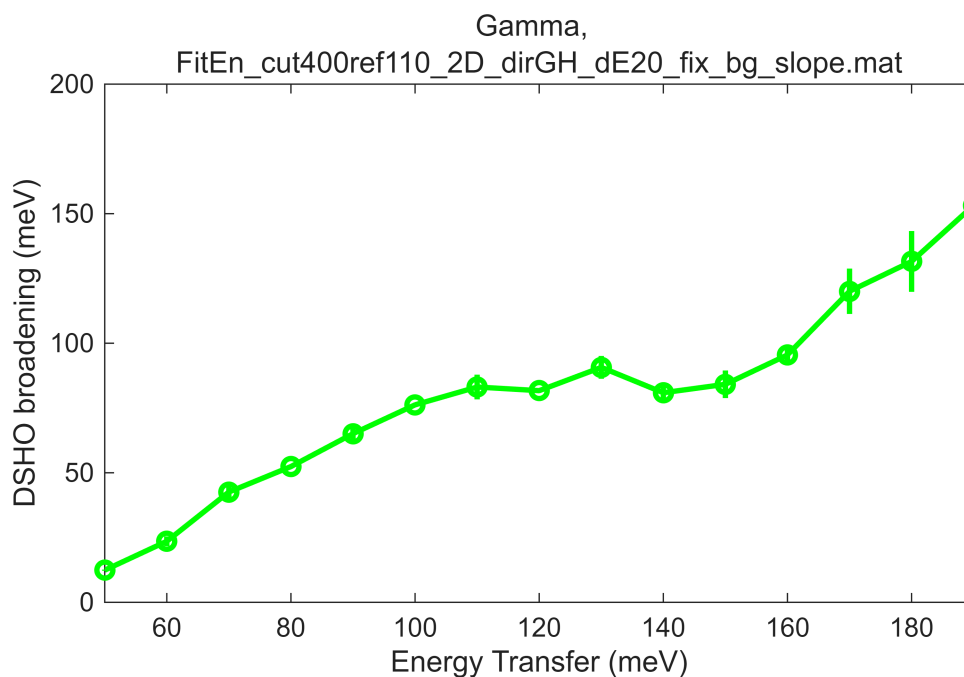
```



```
J0 = fit_res.J0;
pd(J0);ly 20 60;keep_figure
```



```
% incorrect units fixture
G = fit_res.gamma;
G.s_axis.units = 'meV';
fit_res.gamma = G;
pd(G);ly 0 200;keep_figure
```



```
dE = min(Sgn1.x(2:end)-Sgn1.x(1:end-1))
```

```
dE =  
10
```

```
e_min = Sgn1.x(1);
```

```
En = 140
```

```
En =  
140
```

```
fit_par = fit_res.all_fit_par(1);  
dE_step = fit_par.en_range(2); %original energy transfer step data were binned  
to. No point in going finer  
half_dE = 0.5*(fit_par.en_range(3)-fit_par.en_range(1));  
en_idx = floor((En-e_min)/dE)+1;  
%the2Dcuts = cuts2fit.cutsS_list;%(1:2);  
the2Dcuts = {cut(cuts2fit.cut110_sel,[0,0.02,2],[,]);%(1:2);
```

```
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept  
*** Step 1 of 1; Read data for 3903665 pixels -- processing data... -----> retained 3865265 pixels  
-----  
Number of points in input object: 5817504  
Fraction of object selected: 67.10% (= 3903665 points) ; Access speed: 1.38GB/sec  
Fraction of object retained: 66.44% (= 3865265 points)  
  
**** Total time in cut(sqw) : 0.8sec Including:  
Data access time fraction: 24.0%; Transf. time frac.: 53.0%; Resulting pix preparation: 22.8%  
-----
```

```
%init_bg_params =  
%cellfun(@(par)[par(1),par(3)],init_bg_params,'UniformOutput',false); % bg  
%in case of 2-D fitting
```

```

%
[fit_obj,fit_par,figa,figb]=fit_single_set(the2Dcuts,En,half_dE,dE_step,init_fg_p
arams,init_bg_params,false)
lfit_par = fit_res.all_fit_par(en_idx);
en_range = lfit_par.en_range;
[fit_obj,fit_par,figa,figb]=refit_singleset_peaks(the2Dcuts,en_range,lfit_par,0.2
,true,false);

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for 188697 pixels -- processing data... -----> included 154181 pixels
-----
Number of points in input object: 525213
Fraction of object selected: 35.93% (= 188697 points) ; Access speed: 0.89GB/sec
Fraction of object retained: 29.36% (= 154181 points)

**** Total time in cut(sqw) : 0.0sec Including:
Data access time fraction: 33.7%; Transf. time frac.: 61.3%; Resulting pix preparation: 0.0%
-----
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 115133 pixels -- processing data... -----> retained 108055 pixels
-----
Number of points in input object: 525213
Fraction of object selected: 21.92% (= 115133 points) ; Access speed: 1.35GB/sec
Fraction of object retained: 20.57% (= 108055 points)

**** Total time in cut(sqw) : 0.0sec Including:
Data access time fraction: 20.7%; Transf. time frac.: 57.6%; Resulting pix preparation: 21.5%
-----
*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for 349668 pixels -- processing data... -----> included 285973 pixels
-----
Number of points in input object: 525213
Fraction of object selected: 66.58% (= 349668 points) ; Access speed: 1.49GB/sec
Fraction of object retained: 54.45% (= 285973 points)

**** Total time in cut(sqw) : 0.0sec Including:
Data access time fraction: 54.4%; Transf. time frac.: 42.9%; Resulting pix preparation: 0.0%
-----
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 147445 pixels -- processing data... -----> retained 139003 pixels
-----
Number of points in input object: 525213
Fraction of object selected: 28.07% (= 147445 points) ; Access speed: 1.38GB/sec
Fraction of object retained: 26.47% (= 139003 points)

**** Total time in cut(sqw) : 0.0sec Including:
Data access time fraction: 27.4%; Transf. time frac.: 46.6%; Resulting pix preparation: 25.9%
-----
Beginning fit (max 20 iterations)-----
Starting point
-----
Total time = 2.8815s Reduced Chi^2 = 0.94938
Free parameter values:
80.89 1.166 35.2 2.903 2.903
-----
Iteration = 1
-----
Total time = 12.5327s Reduced Chi^2 = 0.86578 Levenberg-Marquardt = 0.1
Free parameter values:
75.83 1.128 34.85 3.645 2.809
-----
Iteration = 2
-----
Total time = 22.0879s Reduced Chi^2 = 0.85548 Levenberg-Marquardt = 0.01
Free parameter values:
70.24 1.048 34.42 4.307 3.256

```

```

-----
Iteration = 3
-----
Total time = 31.7178s    Reduced Chi^2 = 0.86133    Levenberg-Marquardt = 0.001
Free parameter values:
    68.21    1.019    34.31    4.609    3.45
Total time = 34.1394s    Reduced Chi^2 = 0.86177    Levenberg-Marquardt = 0.01
Free parameter values:
    68.73    1.027    34.35    4.539    3.401
Total time = 36.5102s    Reduced Chi^2 = 0.85445    Levenberg-Marquardt = 0.1
Free parameter values:
    69.58    1.039    34.41    4.422    3.32
-----
Iteration = 4
-----
Total time = 45.9687s    Reduced Chi^2 = 0.85573    Levenberg-Marquardt = 0.01
Free parameter values:
    68.57    1.024    34.33    4.56    3.425
Total time = 48.3915s    Reduced Chi^2 = 0.86137    Levenberg-Marquardt = 0.1
Free parameter values:
    69.16    1.033    34.38    4.481    3.369
Total time = 50.7585s    Reduced Chi^2 = 0.85769    Levenberg-Marquardt = 1
Free parameter values:
    69.45    1.037    34.4    4.441    3.339
Total time = 53.2044s    Reduced Chi^2 = 0.85835    Levenberg-Marquardt = 100
Free parameter values:
    69.57    1.039    34.41    4.424    3.322
Total time = 55.5736s    Reduced Chi^2 = 0.86259    Levenberg-Marquardt = 10000
Free parameter values:
    69.58    1.039    34.41    4.422    3.321
Total time = 57.9869s    Reduced Chi^2 = 0.865    Levenberg-Marquardt = 1000000
Free parameter values:
    69.58    1.039    34.41    4.422    3.32
*** No improvement in chi-squared over previous iteration ***
Total time = 57.9946s    Reduced Chi^2 = 0.85445
Free parameter values:
    69.58    1.039    34.41    4.422    3.32
-----

```

Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

```

-----
1          1
2          8
3      69.58 +/- 4.081
4      1.039 +/- 0.04558
5          0
6      34.41 +/- 0.3645
7          0
8          0
9          0
10         0

```

Background parameter values:

```

-----
Function 1:
1      4.422 +/- 0.2997
2     -0.03392
3     -0.009939
4     -0.00194

Function 2:
1      3.32 +/- 0.2158
2     -0.03392

```

3 -0.009939
4 -0.00194

Covariance matrix for free parameters:

```
-----
1.0000    0.9714    0.8301   -0.8233   -0.8345
0.9714    1.0000    0.8132   -0.8393   -0.8087
0.8301    0.8132    1.0000   -0.7608   -0.6674
-0.8233   -0.8393   -0.7608    1.0000    0.3906
-0.8345   -0.8087   -0.6674    0.3906    1.0000
-----
```

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for 525213 pixels -- processing data... -----> included 436622 pixels

Number of points in input object: 525213
Fraction of object selected: 100.00% (= 525213 points) ; Access speed: 1.61GB/sec
Fraction of object retained: 83.13% (= 436622 points)

**** Total time in cut(sqw) : 0.0sec Including:
Data access time fraction: 54.2%; Transf. time frac.: 40.3%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for 108055 pixels -- processing data... -----> included 89204 pixels

Number of points in input object: 108055
Fraction of object selected: 100.00% (= 108055 points) ; Access speed: 2.16GB/sec
Fraction of object retained: 82.55% (= 89204 points)

**** Total time in cut(sqw) : 0.0sec Including:
Data access time fraction: 31.8%; Transf. time frac.: 63.0%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for 139003 pixels -- processing data... -----> included 115365 pixels

Number of points in input object: 139003
Fraction of object selected: 100.00% (= 139003 points) ; Access speed: 1.96GB/sec
Fraction of object retained: 82.99% (= 115365 points)

**** Total time in cut(sqw) : 0.0sec Including:
Data access time fraction: 43.4%; Transf. time frac.: 51.9%; Resulting pix preparation: 0.0%

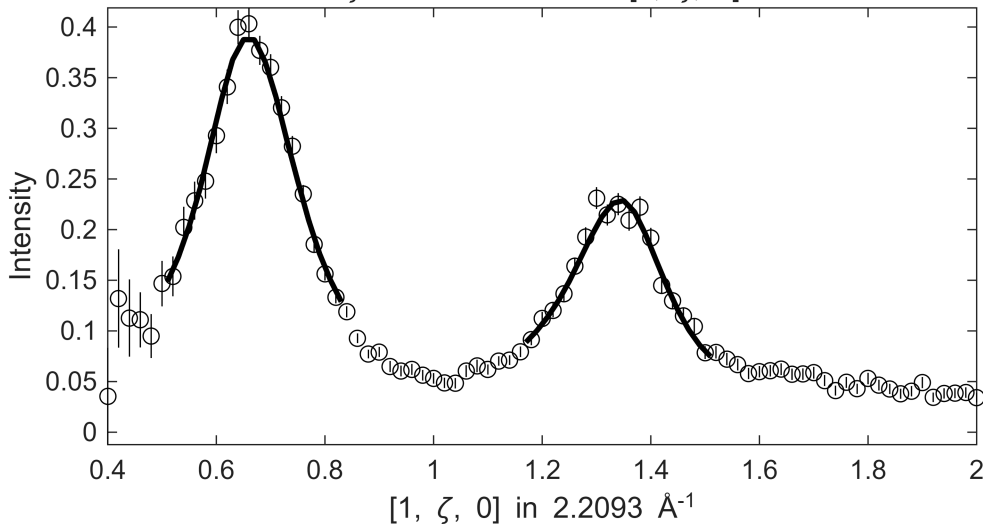
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1

Fe_ei401_align_sym3.sqw

S= 1±0.046; J0= 34±0.36; gamma=69.5815±4.08099

-0.1 ≤ ξ ≤ 0.1 in [1-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [1, 0, η] , 130 ≤ E ≤ 150

ζ=-0.01:0.02:2.01 in [1, ζ, 0]



of 1; Read data for 525213 pixels -- processing data... -----> retained 436622 pixels

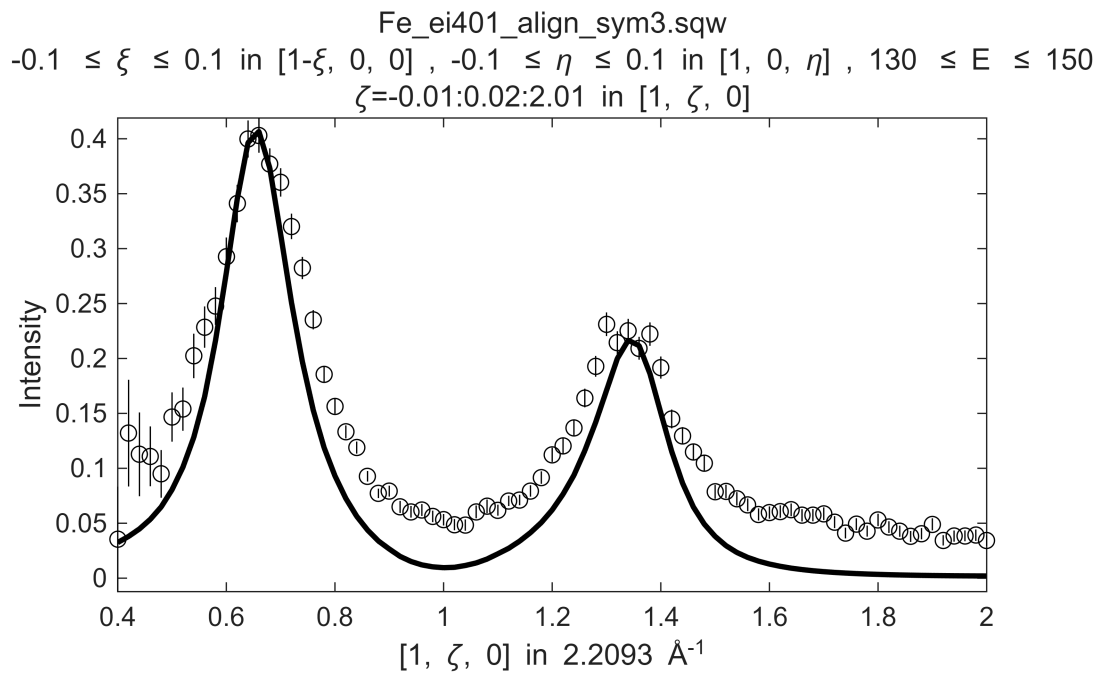
Number of points in input object: 525213

Fraction of object selected: 100.00% (= 525213 points) ; Access speed: 2.12GB/sec

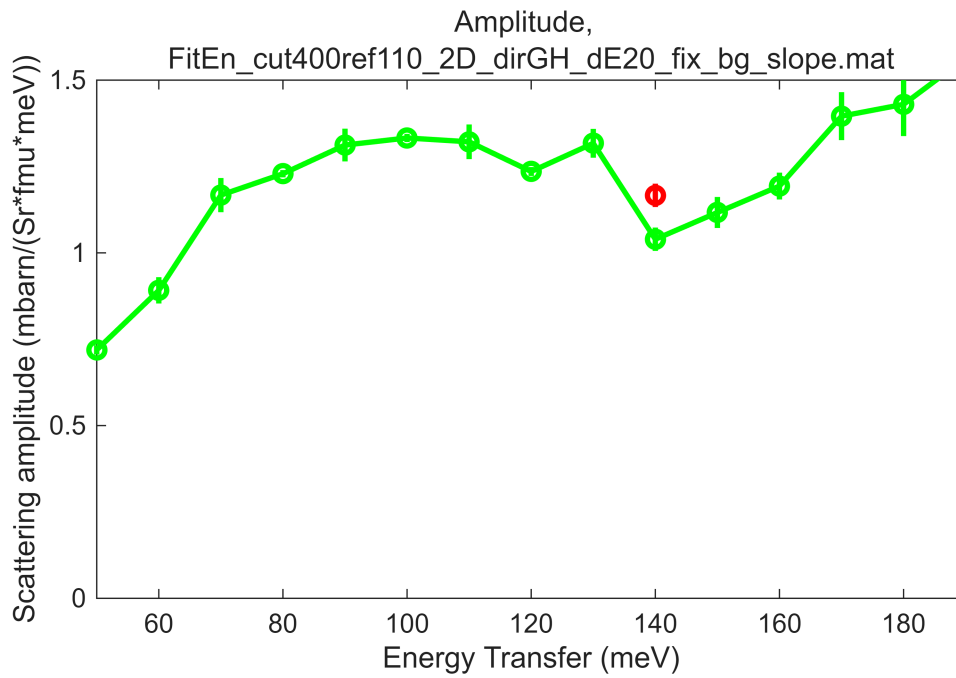
Fraction of object retained: 83.13% (= 436622 points)

**** Total time in cut(sqw) : 0.1sec Including:

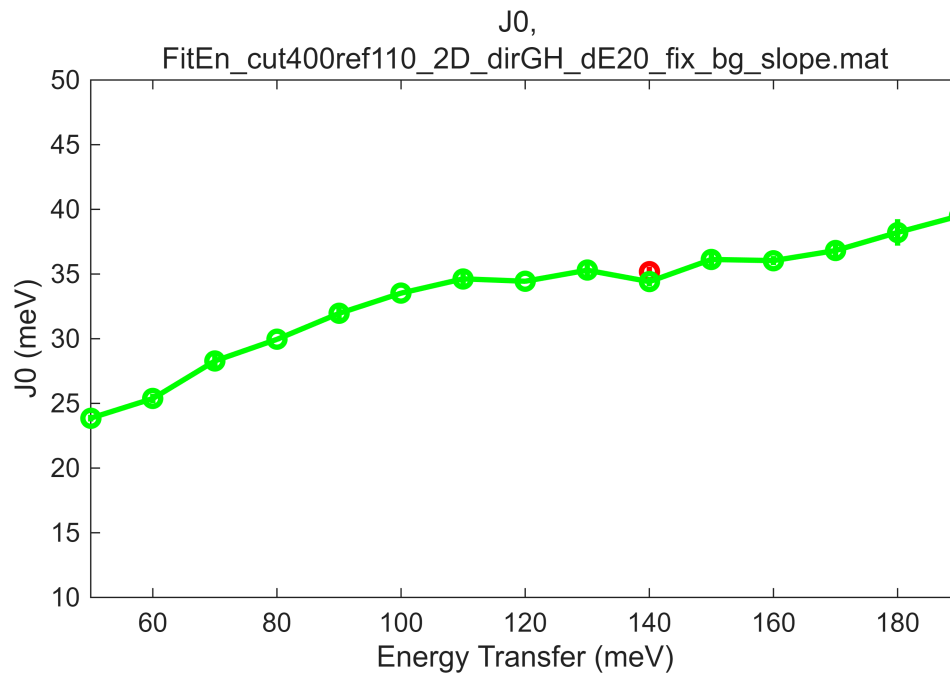
Data access time fraction: 25.9%; Transf. time frac.: 42.7%; Resulting pix preparation: 30.5%



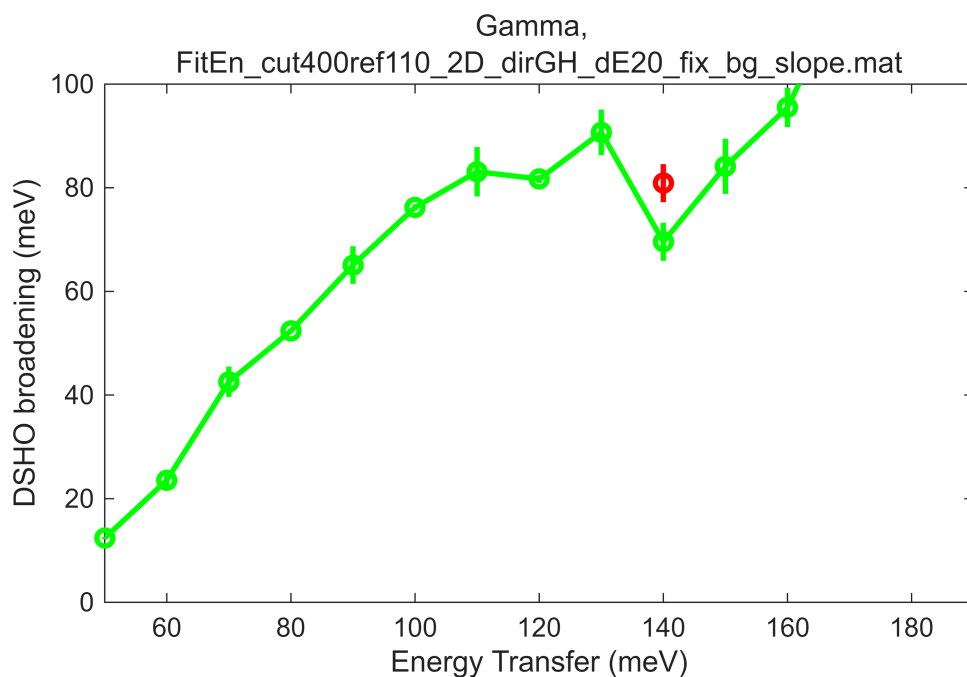
```
%fit_res.all_fit_par(en_idx) = fit_par;  
%fit_res.all_fit_par(en_idx).p = fit_par.p;  
acolor r  
plot(fit_res.S);ly 0 1.5;liny;  
acolor g  
Sgn1.signal(en_idx) = fit_par.p(4);  
pd(Sgn1);ly 0 1.5;liny;keep_figure
```



```
acolor r
plot(fit_res.J0);ly 10 50;liny;
acolor g
J0.signal(en_idx) = fit_par.p(6);
pd(J0);ly 10 50;keep_figure
```



```
acolor r
plot(fit_res.gamma);ly 0 100;liny;
acolor g
G.signal(en_idx) = fit_par.p(3);
pd(G);ly 0 100;keep_figure
```



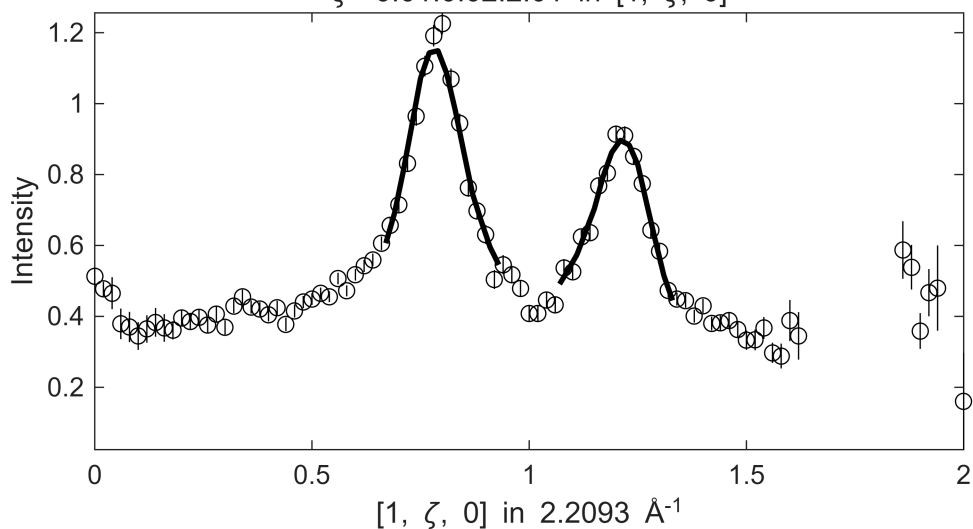
```
[fit_res,Sg,J0,G] = refit_data_around_peaks(the2Dcuts,fit_res,0.2,true);
```

```
*****
```

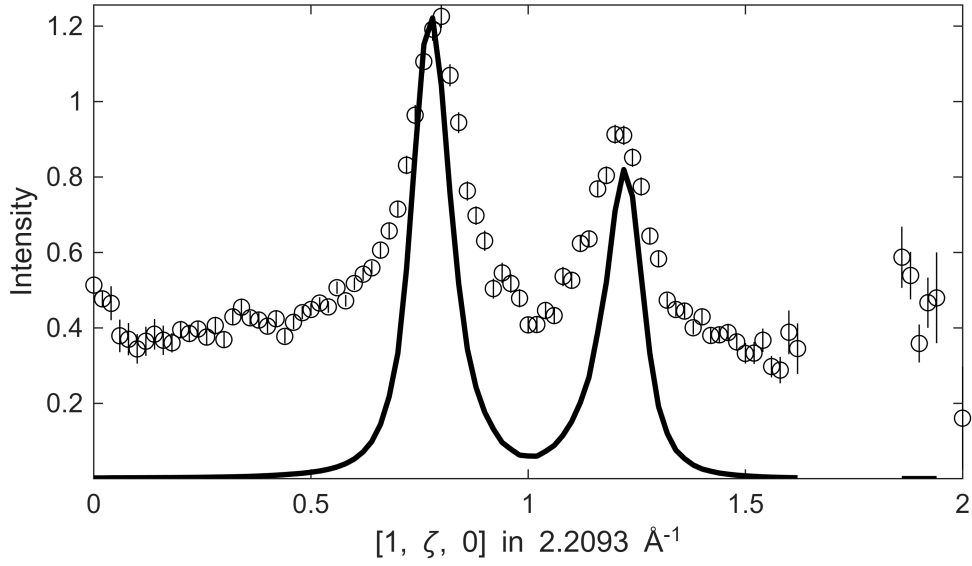
```
*** step: 1#15
```

```
*****
```

Fe_ei401_align_sym3.sqw
 $S = 1.1 \pm 0.052$; $J_0 = 25 \pm 0.2$; $\gamma = 19.4265 \pm 0.900481$
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $40 \leq E \leq 60$
 $\zeta = -0.01:0.02:2.01$ in $[1, \zeta, 0]$



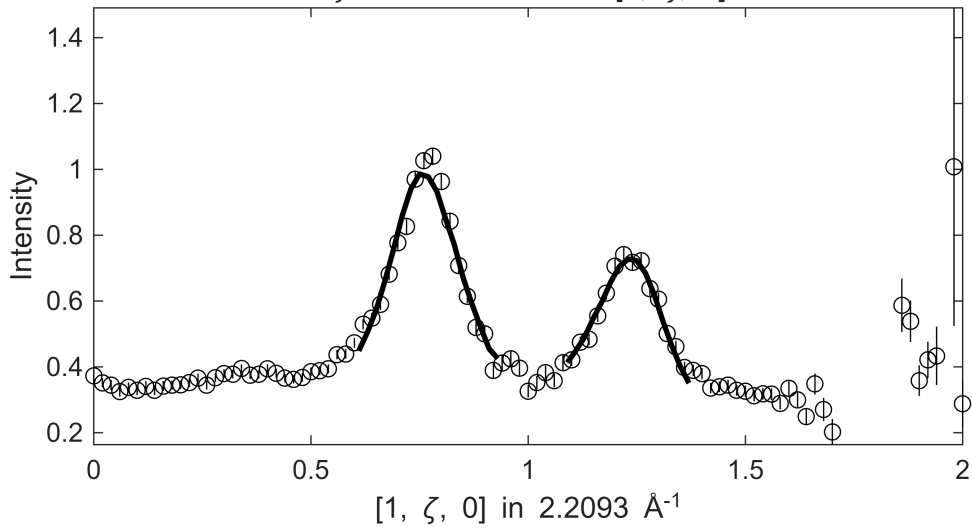
Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $40 \leq E \leq 60$
 $\zeta = -0.01:0.02:2.01$ in $[1, \zeta, 0]$



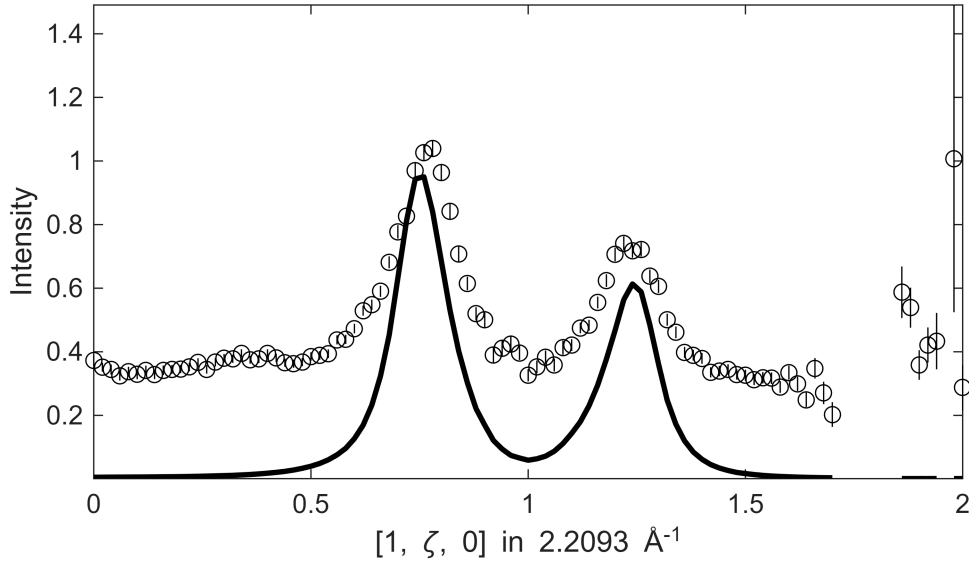
*** step: 2#15

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.16502

Fe_ei401_align_sym3.sqw
 $S = 1.5 \pm 0.076$; $J_0 = 28 \pm 0.3$; $\gamma = 38.704 \pm 1.81518$
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $50 \leq E \leq 70$
 $\zeta = -0.01:0.02:2.01$ in $[1, \zeta, 0]$



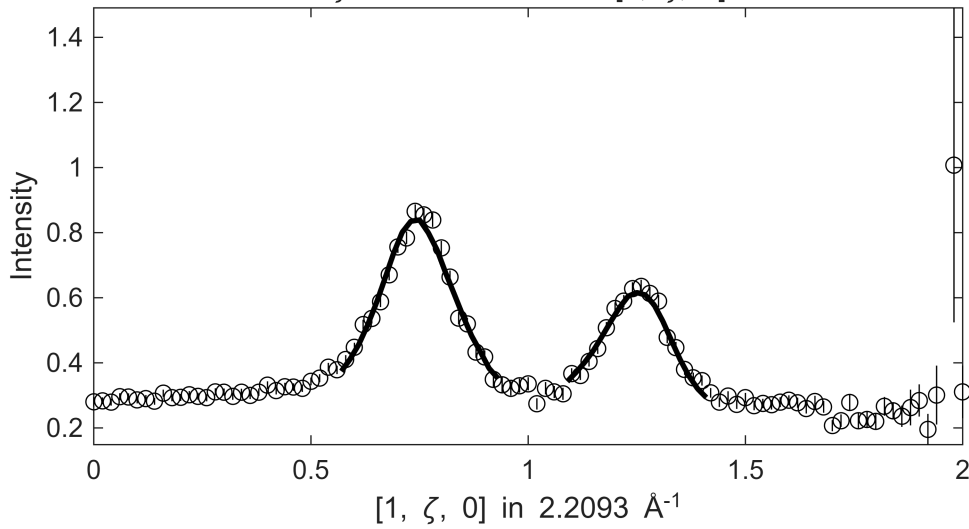
Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $50 \leq E \leq 70$
 $\zeta = -0.01:0.02:2.01$ in $[1, \zeta, 0]$



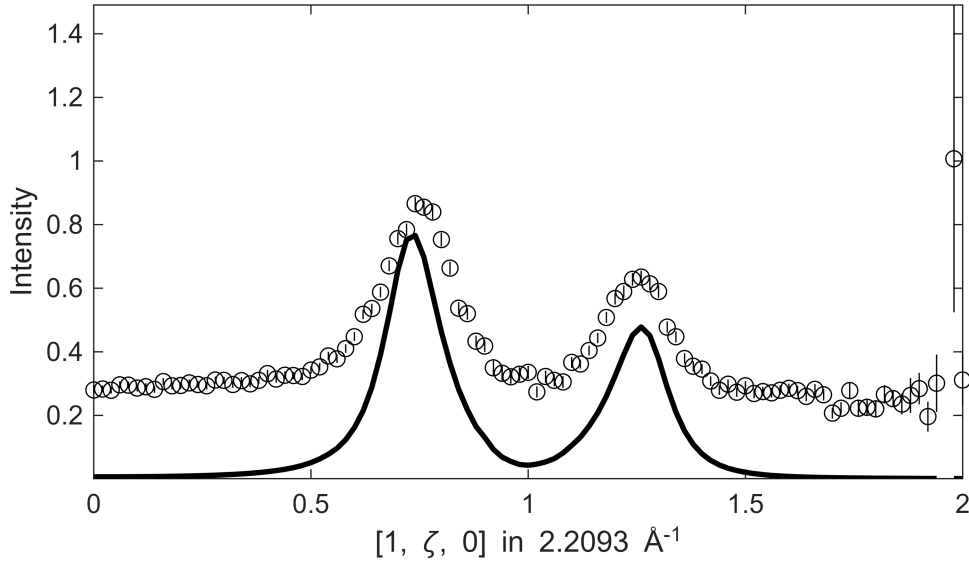
*** step: 3#15

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.16502

Fe_ei401_align_sym3.sqw
 $S = 1.5 \pm 0.11$; $J_0 = 30 \pm 0.75$; $\gamma = 49.8684 \pm 4.8308$
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $60 \leq E \leq 80$
 $\zeta = -0.01:0.02:2.01$ in $[1, \zeta, 0]$



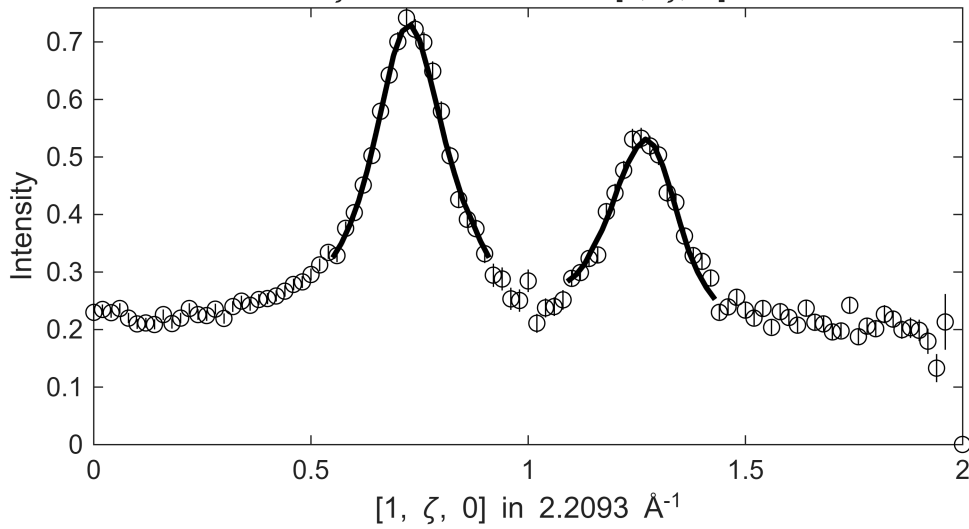
Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $60 \leq E \leq 80$
 $\zeta = -0.01:0.02:2.01$ in $[1, \zeta, 0]$



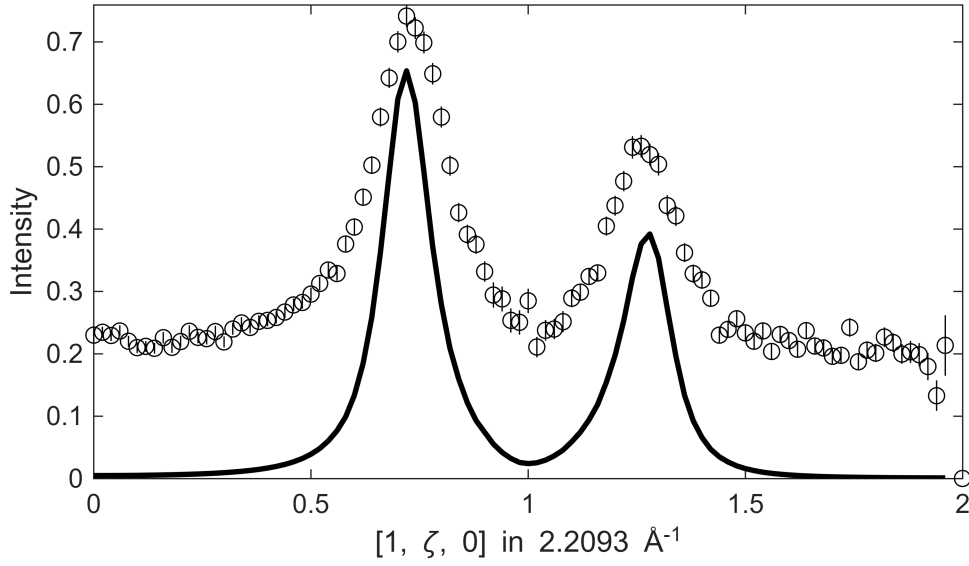
 *** step: 4#15

Warning: Dataset:2 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.33003

Fe_ei401_align_sym3.sqw
 $S = 1 \pm 0.029$; $J0 = 29 \pm 0.16$; $\gamma = 40.0586 \pm 0.853358$
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $70 \leq E \leq 90$
 $\zeta = -0.01:0.02:2.01$ in $[1, \zeta, 0]$



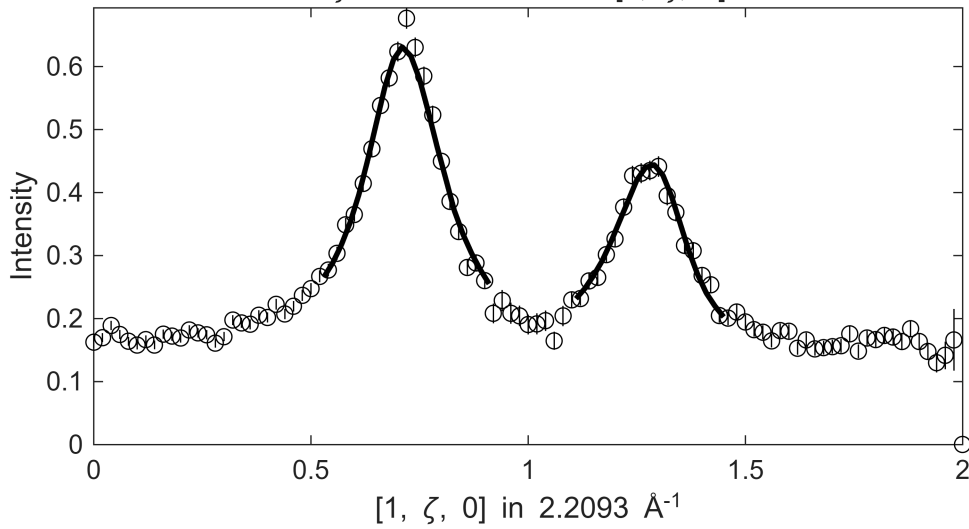
Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $70 \leq E \leq 90$
 $\zeta = -0.01:0.02:2.01$ in $[1, \zeta, 0]$



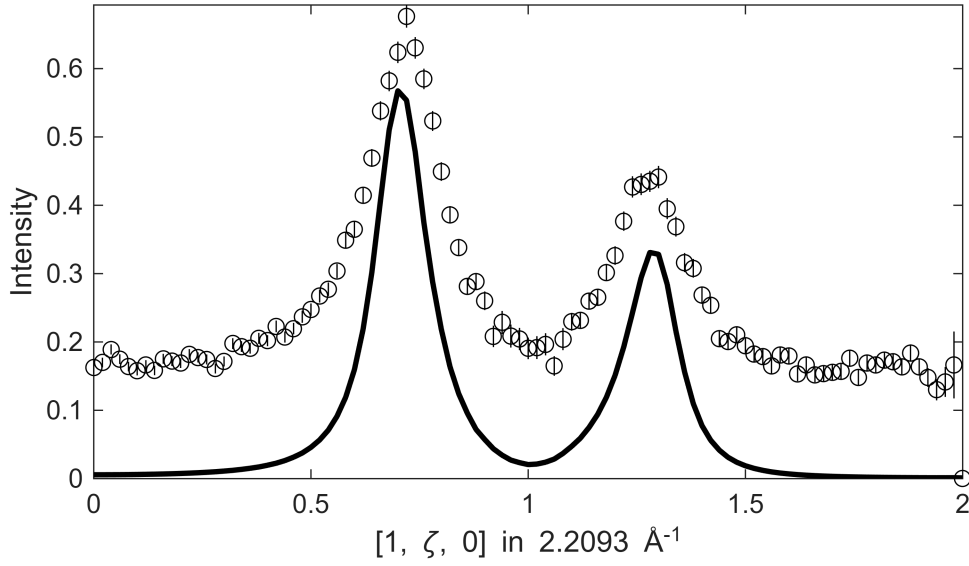
*** step: 5#15

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.16502

Fe_ei401_align_sym3.sqw
 $S = 1 \pm 0.063$; $J_0 = 30 \pm 0.46$; $\gamma = 46.3918 \pm 3.56825$
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $80 \leq E \leq 100$
 $\zeta = -0.01:0.02:2.01$ in $[1, \zeta, 0]$



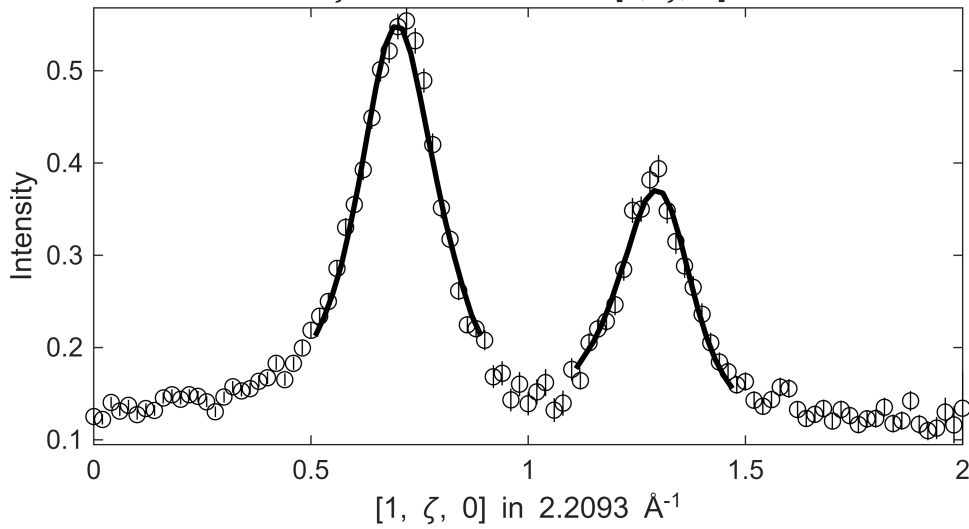
Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $80 \leq E \leq 100$
 $\zeta = -0.01:0.02:2.01$ in $[1, \zeta, 0]$



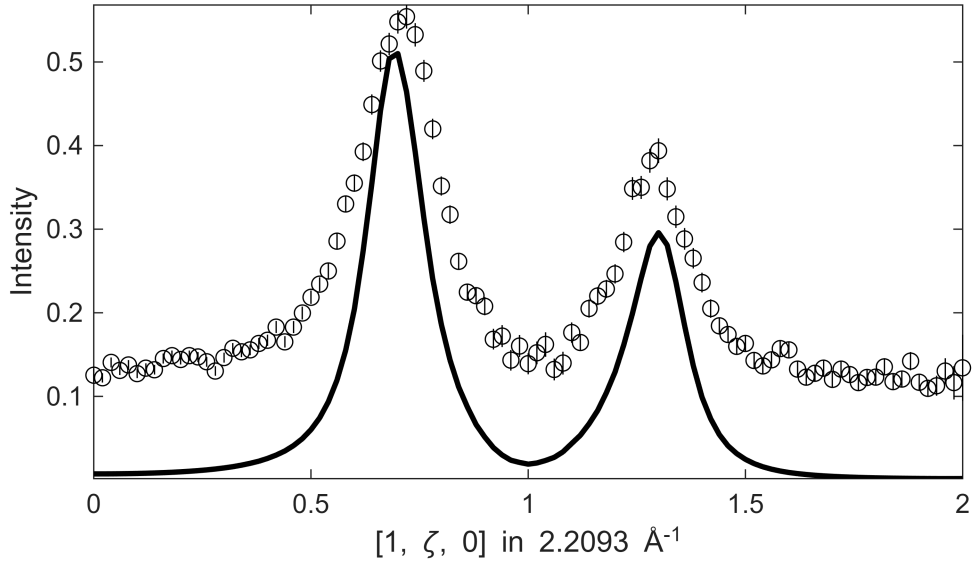
*** step: 6#15

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.16502

Fe_ei401_align_sym3.sqw
 $S = 1.1 \pm 0.069$; $J_0 = 32 \pm 0.54$; $\gamma = 57.7491 \pm 4.32881$
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $90 \leq E \leq 110$
 $\zeta = -0.01:0.02:2.01$ in $[1, \zeta, 0]$

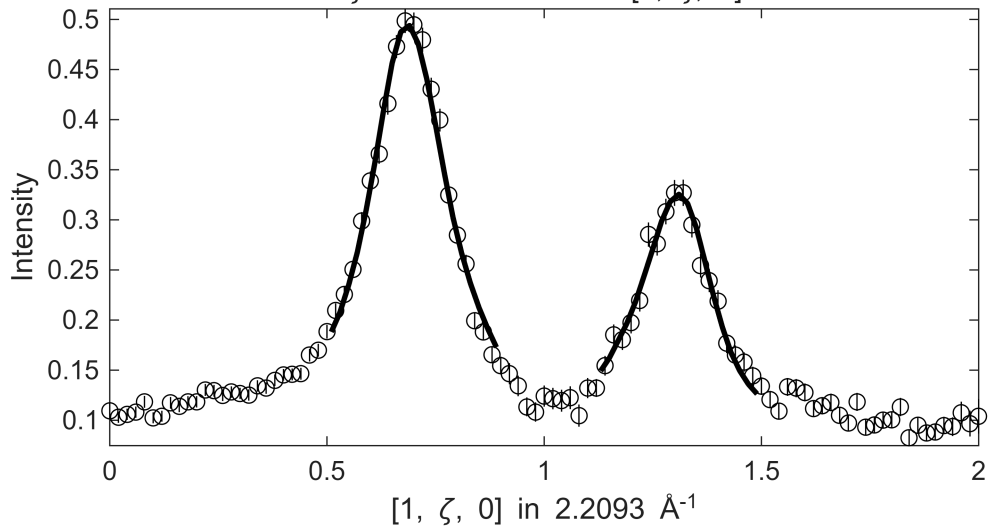


Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $90 \leq E \leq 110$
 $\zeta = -0.01:0.02:2.01$ in $[1, \zeta, 0]$

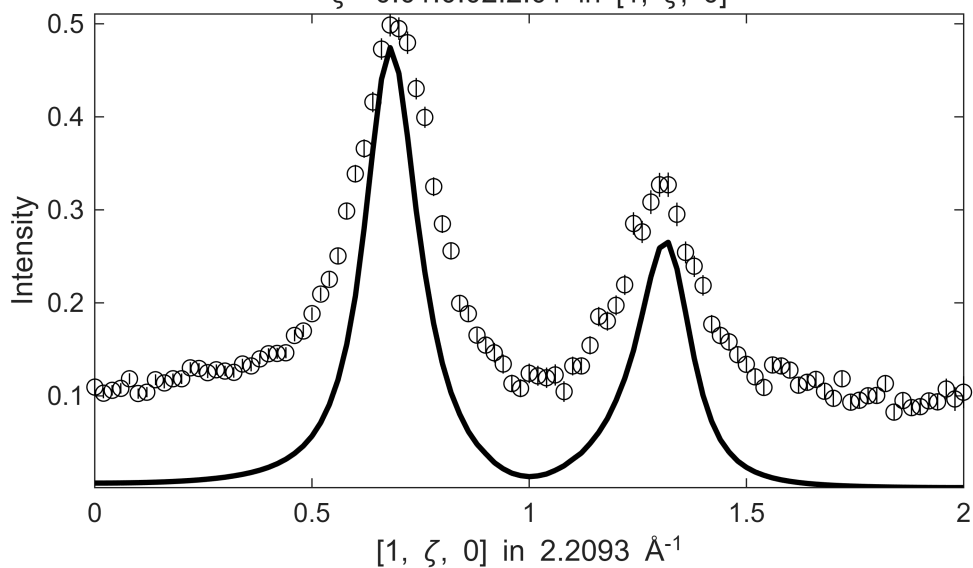


 *** step: 7#15

Fe_ei401_align_sym3.sqw
 $S=0.96 \pm 0.04$; $J_0 = 32 \pm 0.26$; $\gamma = 53.483 \pm 2.20037$
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $100 \leq E \leq 120$
 $\zeta = -0.01:0.02:2.01$ in $[1, \zeta, 0]$

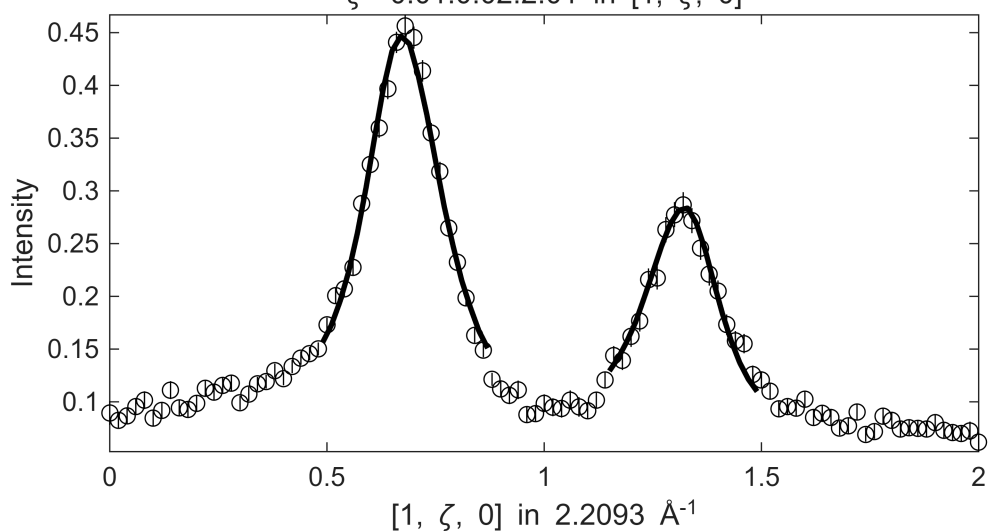


Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $100 \leq E \leq 120$
 $\zeta = -0.01:0.02:2.01$ in $[1, \zeta, 0]$

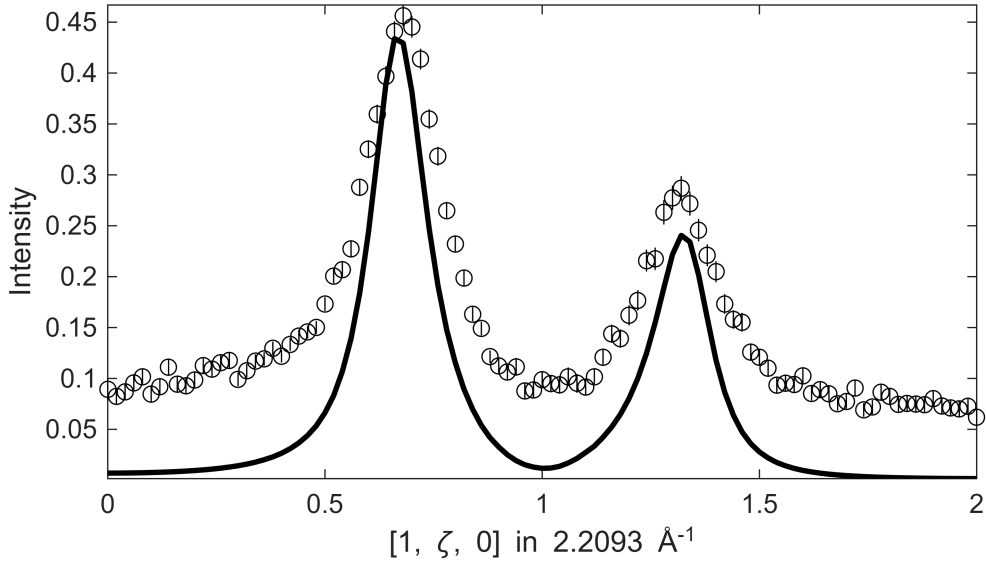


*** step: 8#15

Fe_ei401_align_sym3.sqw
 $S=0.98 \pm 0.049$; $J_0 = 33 \pm 0.37$; $\gamma = 59.994 \pm 3.56646$
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $110 \leq E \leq 130$
 $\zeta = -0.01:0.02:2.01$ in $[1, \zeta, 0]$



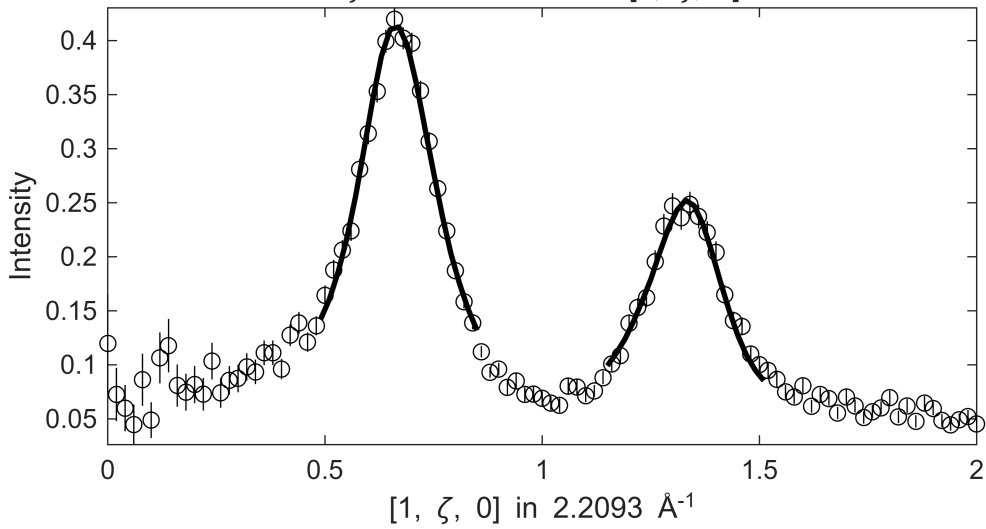
Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $110 \leq E \leq 130$
 $\zeta = -0.01:0.02:2.01$ in $[1, \zeta, 0]$



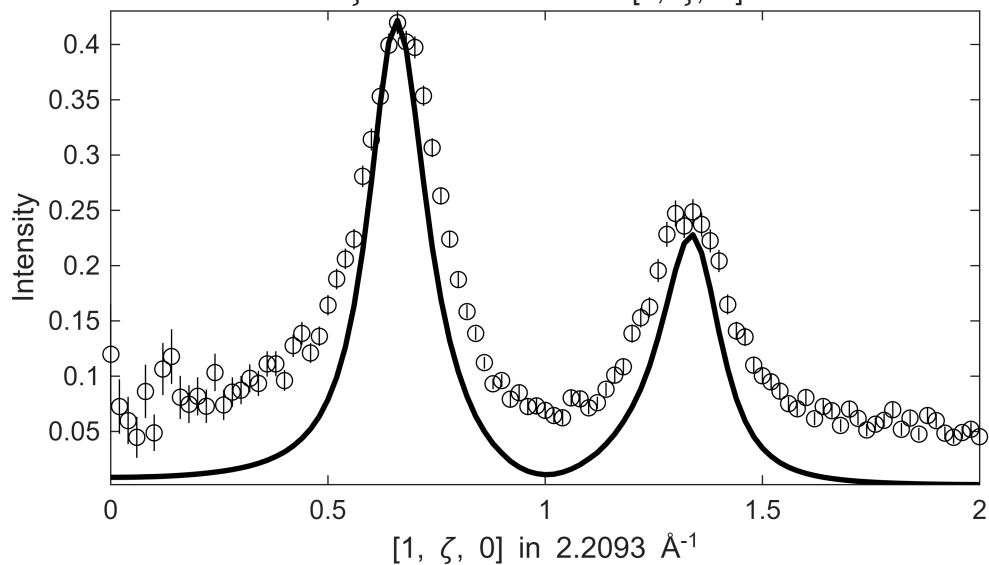
*** step: 9#15

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.16502

Fe_ei401_align_sym3.sqw
 $S = 1 \pm 0.054$; $J_0 = 34 \pm 0.41$; $\gamma = 67.8742 \pm 4.3168$
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $120 \leq E \leq 140$
 $\zeta = -0.01:0.02:2.01$ in $[1, \zeta, 0]$

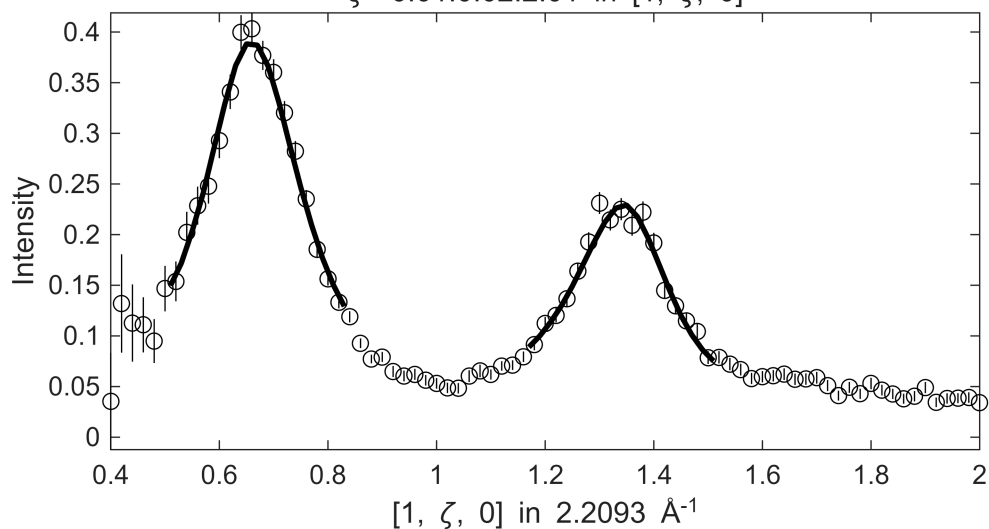


Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $120 \leq E \leq 140$
 $\zeta = -0.01:0.02:2.01$ in $[1, \zeta, 0]$

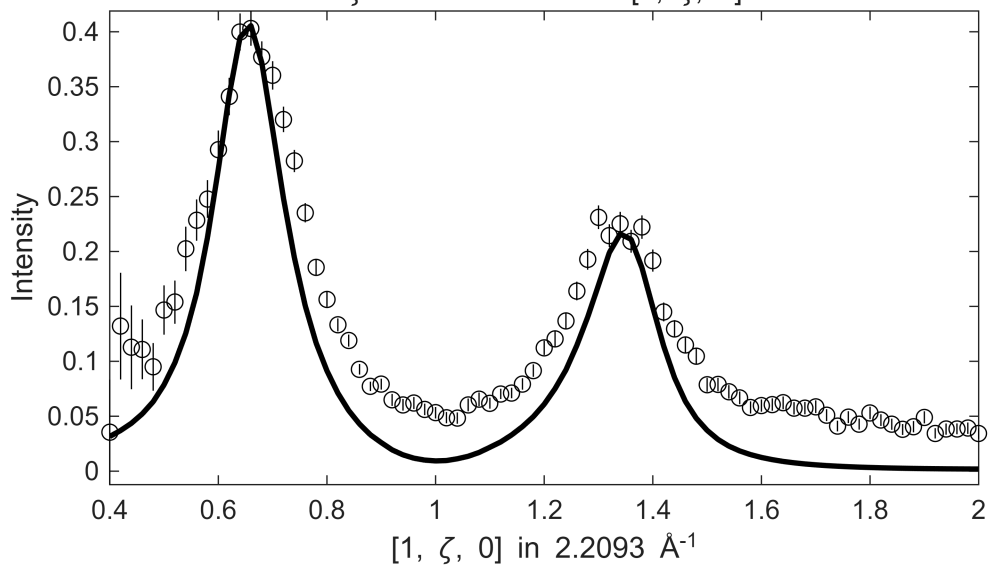


*** step: 10#15

Fe_ei401_align_sym3.sqw
 $S = 1 \pm 0.046$; $J_0 = 34 \pm 0.36$; $\gamma = 68.2867 \pm 4.09479$
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $130 \leq E \leq 150$
 $\zeta = -0.01:0.02:2.01$ in $[1, \zeta, 0]$



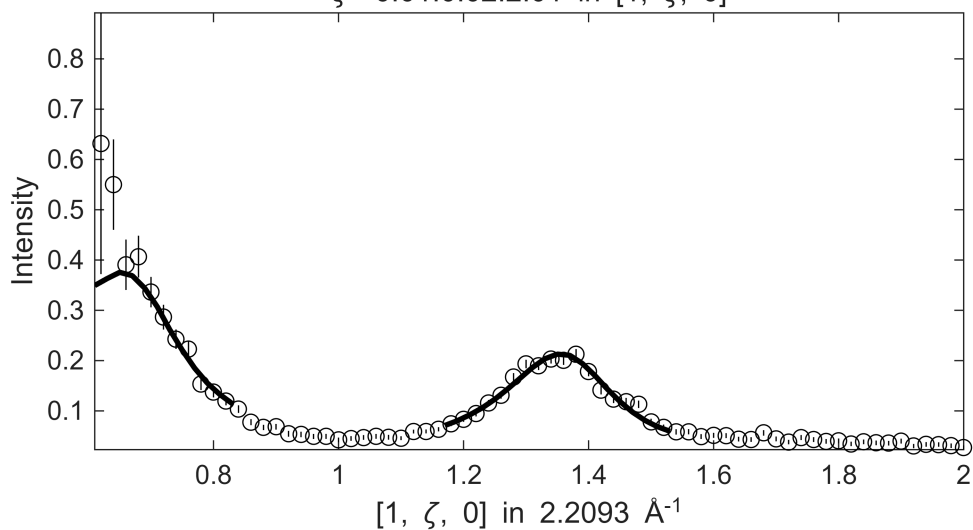
Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $130 \leq E \leq 150$
 $\zeta = -0.01:0.02:2.01$ in $[1, \zeta, 0]$



*** step: 11#15

Warning: Dataset:2 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.33003

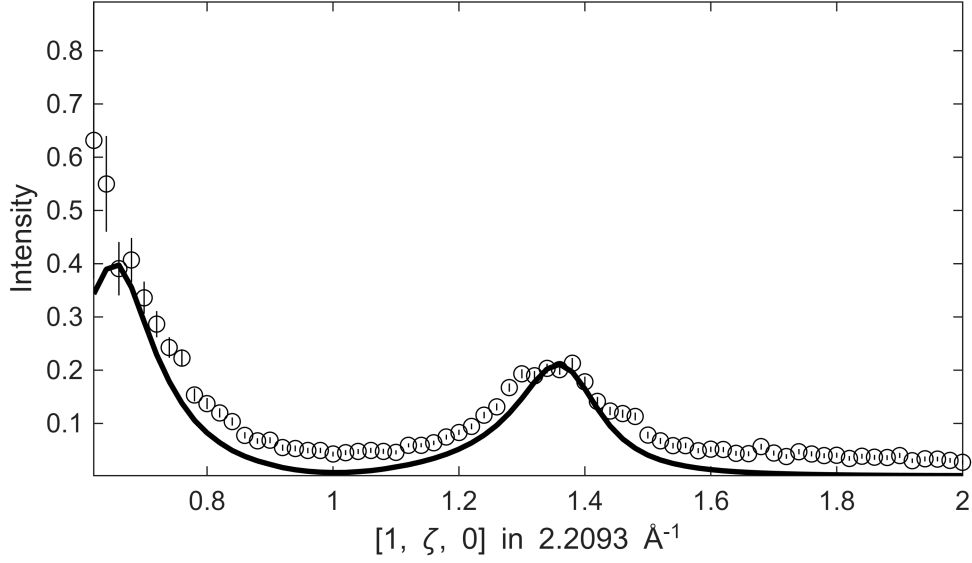
Fe_ei401_align_sym3.sqw
 $S = 1 \pm 0.053$; $J_0 = 35 \pm 0.41$; $\gamma = 67.3789 \pm 4.94554$
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $140 \leq E \leq 160$
 $\zeta = -0.01:0.02:2.01$ in $[1, \zeta, 0]$



Fe_ei401_align_sym3.sqw

$-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $140 \leq E \leq 160$

$\zeta=-0.01:0.02:2.01$ in $[1, \zeta, 0]$



*** step: 12#15

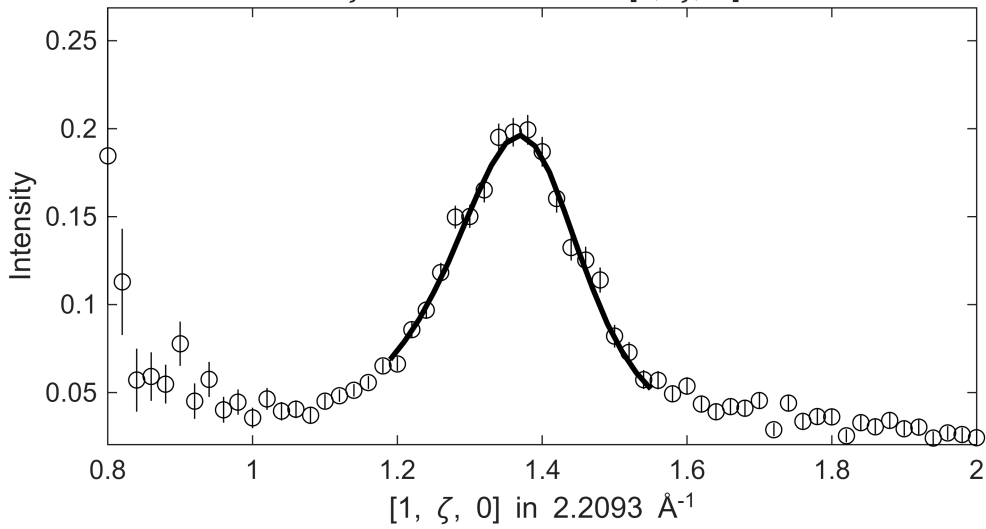
Warning: Dataset:3 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.49505

Fe_ei401_align_sym3.sqw

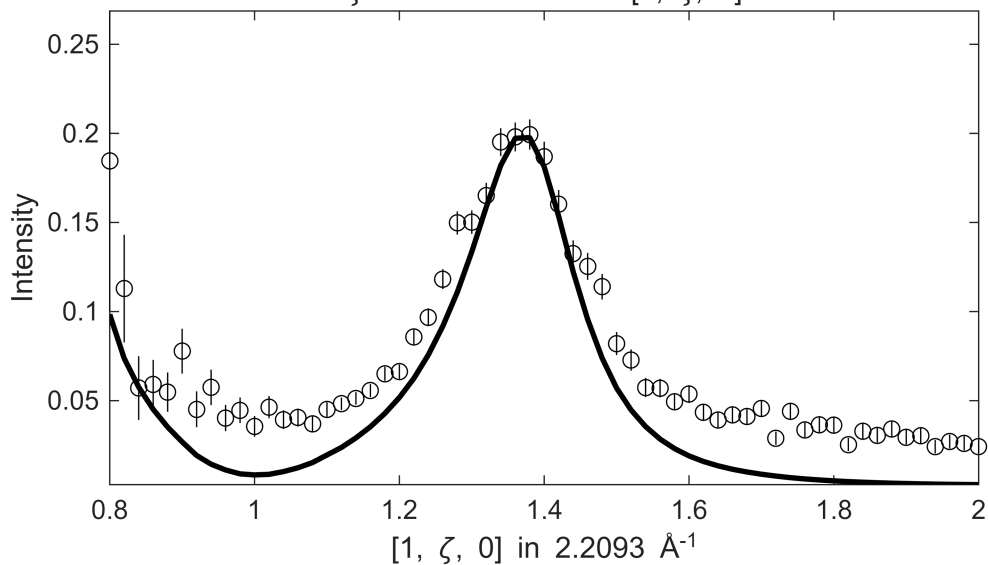
$S= 1.2 \pm 0.1$; $J_0= 35 \pm 0.54$; $\gamma=82.5657 \pm 7.7325$

$-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $150 \leq E \leq 170$

$\zeta=-0.01:0.02:2.01$ in $[1, \zeta, 0]$



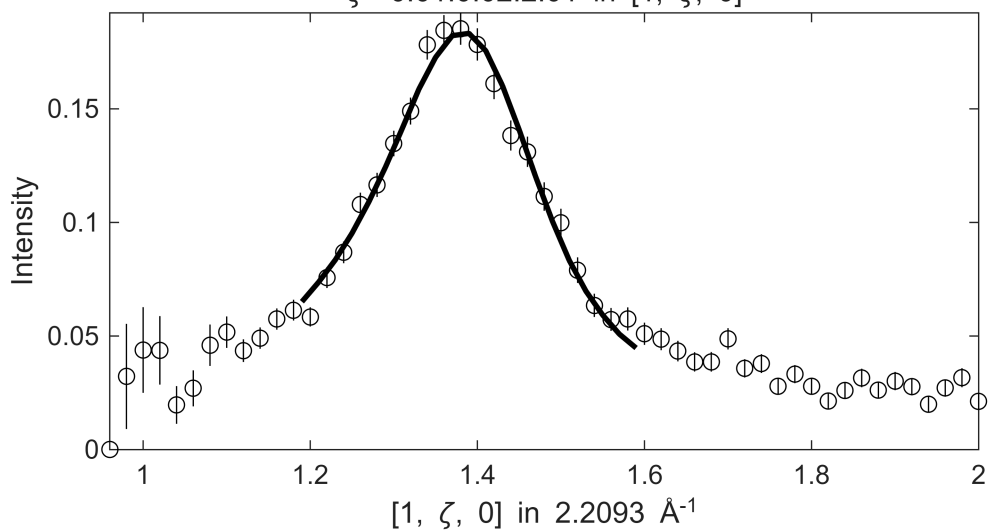
Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $150 \leq E \leq 170$
 $\zeta = -0.01:0.02:2.01$ in $[1, \zeta, 0]$



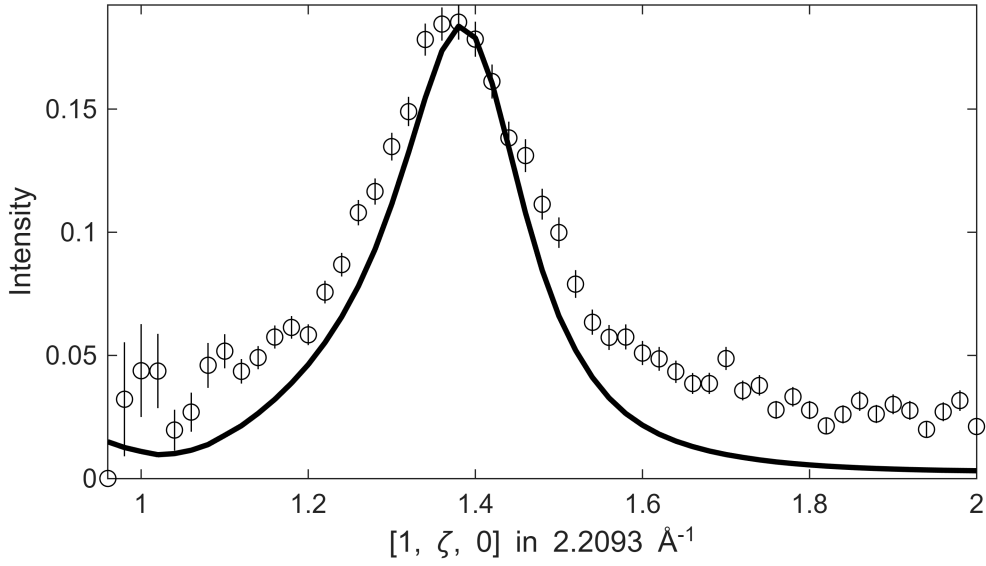
*** step: 13#15

Warning: Dataset:4 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.66007

Fe_ei401_align_sym3.sqw
 $S = 1.1 \pm 0.065$; $J0 = 35 \pm 0.5$; $\gamma = 86.04 \pm 6.60787$
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $160 \leq E \leq 180$
 $\zeta = -0.01:0.02:2.01$ in $[1, \zeta, 0]$



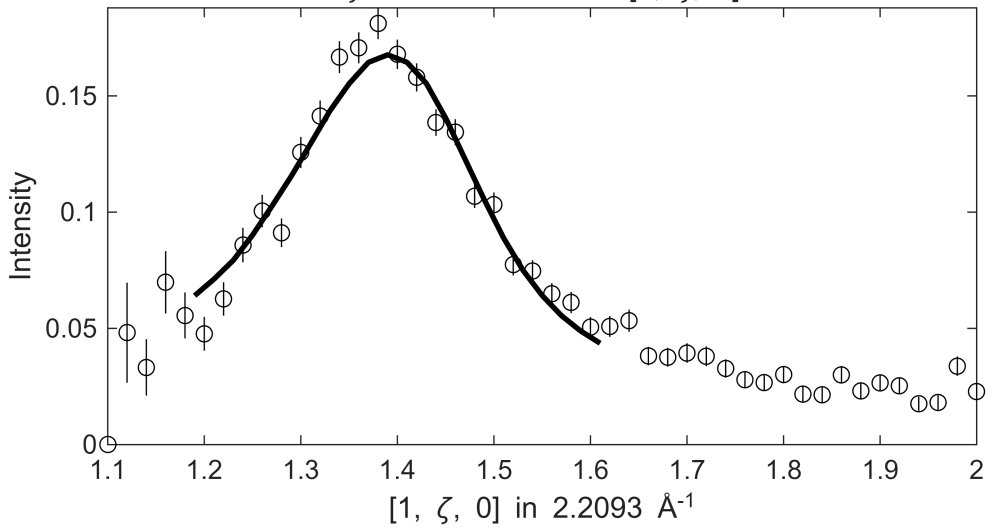
Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $160 \leq E \leq 180$
 $\zeta = -0.01:0.02:2.01$ in $[1, \zeta, 0]$



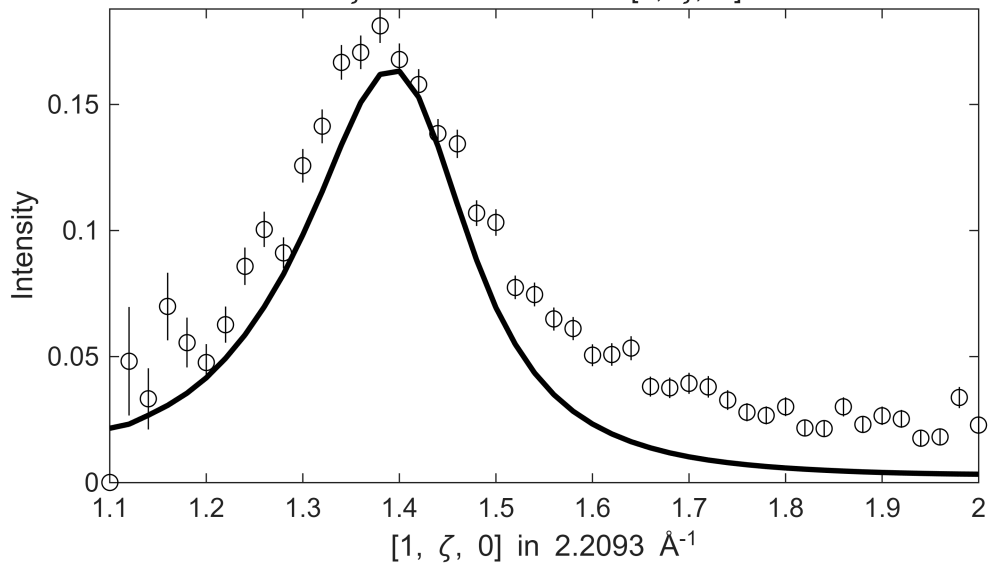
*** step: 14#15

Warning: Dataset:2 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.33003

Fe_ei401_align_sym3.sqw
 $S = 1.1 \pm 0.11$; $J0 = 36 \pm 0.73$; $\gamma = 90.9606 \pm 9.88504$
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $170 \leq E \leq 190$
 $\zeta = -0.01:0.02:2.01$ in $[1, \zeta, 0]$



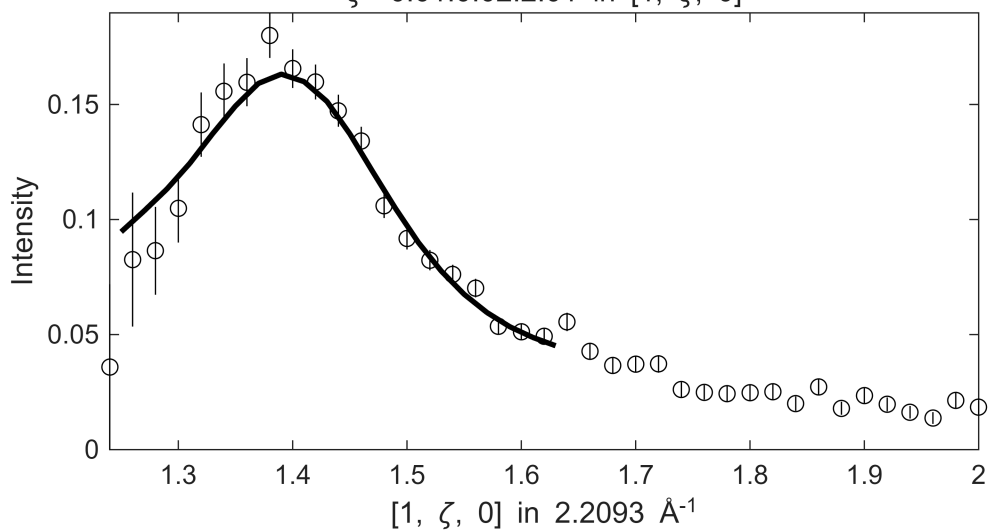
Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $170 \leq E \leq 190$
 $\zeta = -0.01:0.02:2.01$ in $[1, \zeta, 0]$



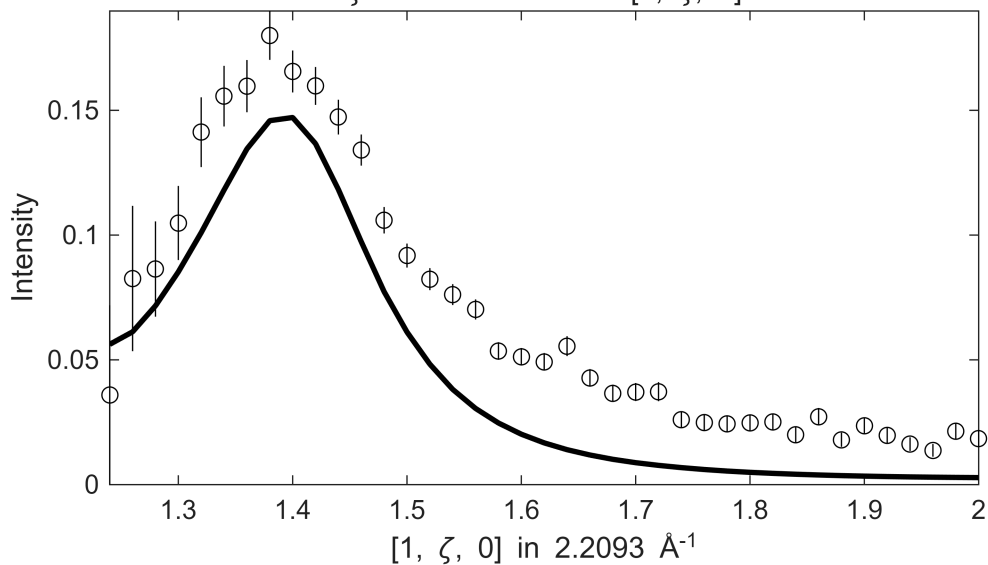
*** step: 15#15

Warning: Dataset:2 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.33003

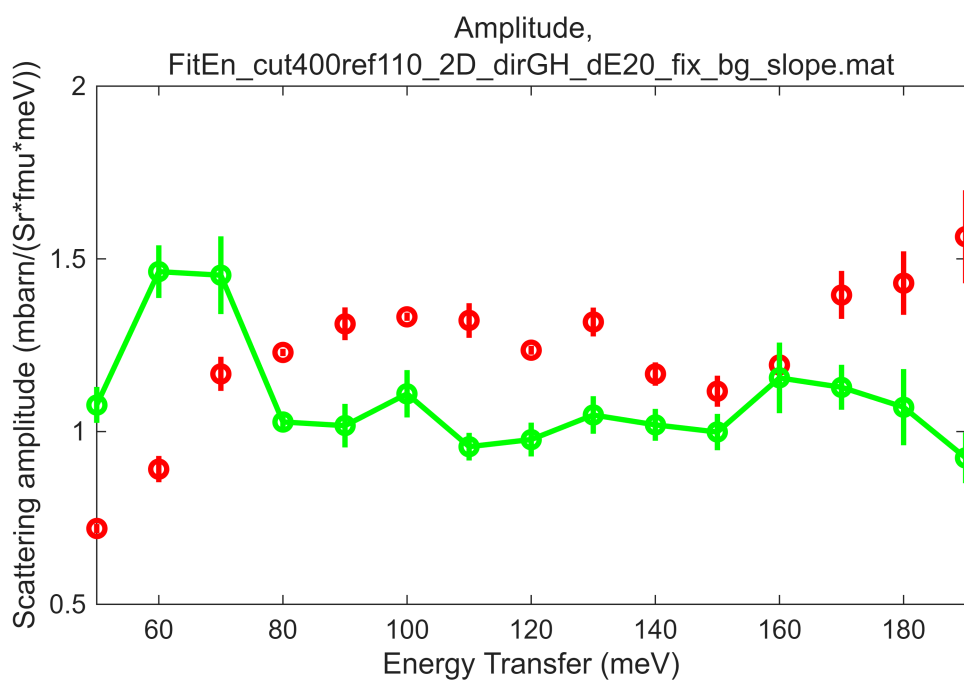
Fe_ei401_align_sym3.sqw
 $S=0.92 \pm 0.073$; $J_0 = 36 \pm 0.36$; $\gamma = 86.7402 \pm 5.71139$
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $180 \leq E \leq 200$
 $\zeta = -0.01:0.02:2.01$ in $[1, \zeta, 0]$



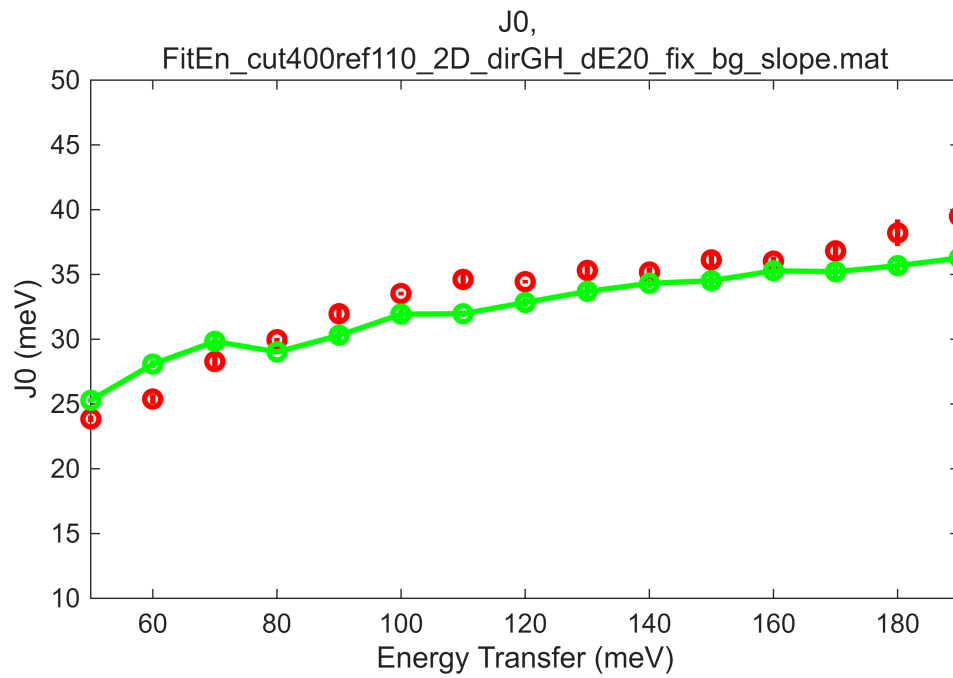
Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $180 \leq E \leq 200$
 $\zeta = -0.01:0.02:2.01$ in $[1, \zeta, 0]$



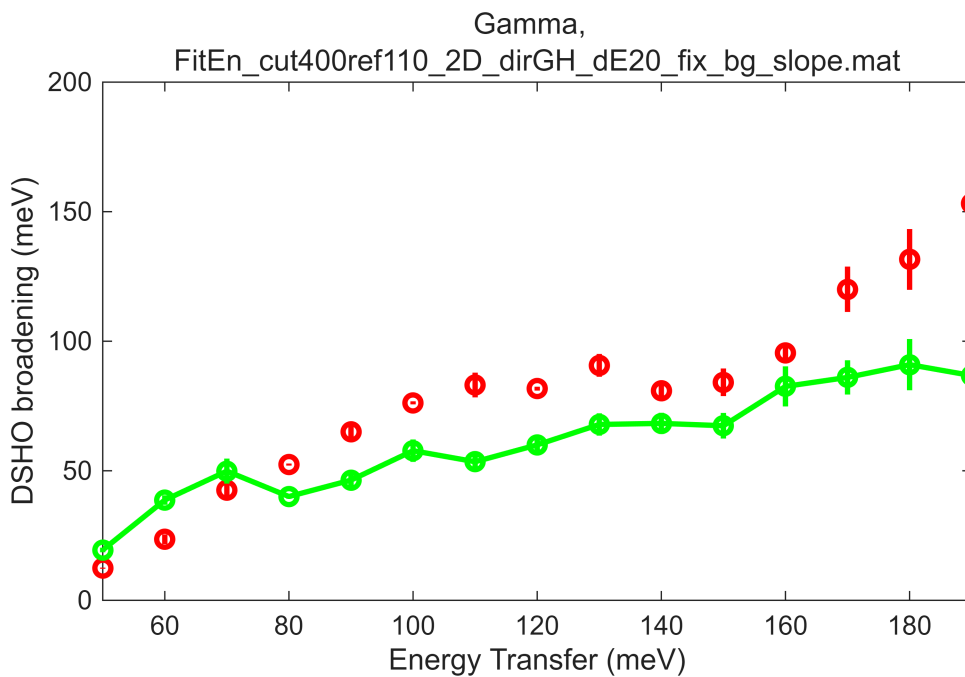
```
acolor r
plot(fit_res.S);liny;
acolor g
pd(Sg);ly 0.5 2;liny;keep_figure
```



```
acolor r
plot(fit_res.J0);liny;
acolor g
pd(J0);ly 10 50;keep_figure
```



```
acolor r
plot(fit_res.gamma);liny;
acolor g
pd(G);ly 0 200;keep_figure
```



```
% fit_res.Speak_varsl = Sg;
% fit_res.J0peak_varsl = J0;
% fit_res.gamma_peak_varsl = G;
fit_res.Speak = Sg;
fit_res.J0peak = J0;
```

```
fit_res.gamma_peak = G;  
fit_res.peak_fit_par = [peak_fit_par{:}];  
fit_res.peak_fit_VarSlope_par = fit_res.peak_fit_par;  
fit_resEi400 = fit_res;  
save(fullfile(fit_data_path,fit_data_file),"fit_resEi400");
```